

2. Explain the difference between method declaration and method body.

Method declaration describes the method's name return type and parameters

Method body contain code that runs when it is called

3. What type of keyword is used to change the access level of a method?

Access modifier

4. What is another word used for describing the access level of a method?

visibility

5. Explain the scope of each of the variables in the code below:

```
public class ScopeExample {  
    public static void main(String[] args) {  
        int var1;  
        for (int var2 = 0; var2 < 5; var2++) {  
            method1();  
        }  
    }  
    public static void method1() {  
        int var3;  
        for (int var4 = 0; var4 < 2; var4++) {  
            var3 += 1;  
        }  
    }  
}
```

Each variable is defined in between the curly brackets

Var 1=main method var 2=the loop var3=method1 var4=loop in method1

6. Write a method declaration for each of the following descriptions:

a) A class method named getVowels that can be called by any other method, requires a String parameter, and returns an integer value.

public static int getVowels()

b) A class method named extractDigit that can be called by any other method, requires an integer parameter, and returns an integer value.

private static int extractDigit(int[] intAry)

c) A class method named insertString that can be called by any other method, requires a String parameter and an integer parameter, and returns a String parameter.

public static String insertString(String, int)

7. a) How does the compiler distinguish one method from another?

Method name

b) Can two methods in the same class have the same name? Explain.

yes

8. a) What is the return statement used for?

Ends the method

b) How many values can a return statement send back to the calling statement?

One value

c) How is the declaration of a method returning a value different from the declaration of a method that does not return a value?

A method returning must return a value

A method that doesn't return a value uses void

9. Find and explain the error in the code below: `public class MethodCallExample { public static void main(String[] args) { int num;`

`You call a method that returns a value but not taking its value`